

Question	Answer	Marks
3(a)	$E = \frac{1}{2} m \omega^2 x_0^2$	C1
	$2.2 \times 10^{-4} = \frac{1}{2} \times 24 \times 10^{-3} \times (14 \times 10^{-3} / 4)^2 \times \omega^2$	C1
	$\omega = 39 \text{ rad s}^{-1}$	A1

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3(b)(i)	use of acceleration = 9.81 m s^{-2}	C1
	$x_0 = 9.81 / 39^2$ $= 6.4 \times 10^{-3} \text{ m}$	A1
3(b)(ii)	at top of oscillation	B1
	<i>any one point from:</i> where the downward acceleration first exceeds free-fall acceleration where the greatest downwards acceleration occurs where the resultant force is the maximum downwards where the contact force is a minimum	B1

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